# Pasteables: A Flexible and Smart "Stick-and-Peel" Wearable Platform for Fitness and Athletics

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***Abstract*—Wearable technologies, such as smartwatches and health monitoring bands, are becoming increasingly popular and transforming our fitness and sports industry. Monitoring health parameters and body activity can help achieve fitness goals, improve sports performance, and aid in physiotherapy. However, state-of-the-art wearable devices are often not flexible (i.e., targeted to a specific use case), expensive, hard to setup, or have privacy concerns. They are designed and optimized for a specific application with tight hardware–software integration, and it is hard or impossible to repurpose them for diverse use cases. In this article, we propose a flexible, reconfigurable human body movement and health monitoring platform, called "Pasteables." It is a stick-and-peel device (which acts like a band-aid) that attaches to the skin or clothing and can create an on-body network of smart wearables. Given an application, Pasteables can be easily adapted to its requirements while ensuring good performance and privacy. We present the overall architecture, system design steps, and the configuration process. We have designed a set of functional prototypes and**

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**configured them for performing a variety of cameraless pose and posture detection tasks, which are important for athletes, professional dancing, physiotherapy, and fitness workouts. We note that the Pasteables can perform pose/posture detection without external stationary reference at low cost and high accuracy.**

**Wearable devices, such** as smartwatches and fitness bands, are being increasingly used for fitness and sport performance monitoring. These wearable and health monitoring devices track a small set of health/motion parameters[1] and are permanently configured to be used within one use case, such as detecting incorrect exercise motion, identifying flaws in a dancing student's pose, catching synchronization issues with synchronized swimming or gymnastics, tracking walking posture problems, and monitoring athlete's performance issues. Due to this, multiple wearable devices are needed for each application, which can be costly to the end user wanting to perform all these tasks. The tight coupling between the platform, hardware components, and software application limits possibilities and leads to underutilization of different system components. Therefore, a cost-effective and reconfigurable wearable platform, which can be repurposed for many applications will be highly appreciated in the fitness and sports industry.

In this article, we propose a new paradigm of stick-and-peel, reconfigurable, cost-effective, secure, and user-configurable wearable devices, called Pasteables, which can be employed in diverse fitness/health/sports applications. A set of Pasteables can create an on-body network of sensors for one or more users to collectively perform diverse tasks. Each Pasteable consists of a base pad and pasted-on components, such as a microcontroller called Brain in Pasteable (BIP), power management unit (PMU), peripheral devices (e.g., sensors and actuators), and notification LED array. The BIP communicates with an end device (e.g., smartphone or cloud) for data transmission and decision-making. Peripheral devices are implemented uniformly in a wrapper circuit with a peripheral controller, which can be programmed to communicate using the same interface. With these components, Pasteables can configure its system automatically, so the wearer only focuses on pasting the sensors, determining what data they want to collect, and how to process them for achieving the target functionality. Our Pasteable prototypes are built using a plastic sheet for structural support, copper sheet strips for the interconnections, and skin-safe adhesive to stick the pad to the skin or clothing, such as band-aid.

Pose and posture detection applications are widely used in fitness and athletics monitoring.[2] Camera-based techniques for pose/posture detection compromises user privacy with data-like faces and locations. They also need an external fixed reference for calibration. Available cameraless systems for pose/posture detection are expensive and not reconfigurable.

To address these issues with existing solutions, we present Pasteables and demonstrate their effectiveness and flexibility for various pose/posture detection applications. Specifically, we have configured a set of Pasteables for performing the following three applications.

1) Detecting correct/incorrect fitness exercise motions.
2) Detecting a specific exercise based on the subject's motion.
3) Determining walking posture correctness.

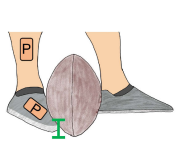
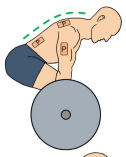
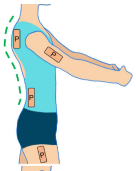
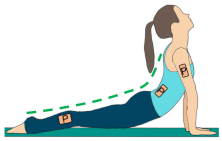
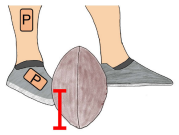
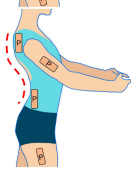
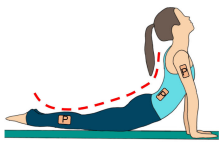
These specific applications can be extended to sports training, fitness workout, dance, physiotherapy, and other fitness applications. With the framework, one can make cost-effective utilization of the same set of base pads and CUIs to realize a diverse set of applications. We verify the functionality, configurability, and effectiveness of Pasteables for these applications through extensive experimental evaluations with the help of multiple volunteers.

In this article, we make the following contributions.

1) We present a stick-and-peel band-aid-like wearable platform, Pasteables, that can be easily adapted to a wide range of applications



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| Application | Sports Training | Fitness Workout | Dance | Physiotherapy |
|---|---|---|---|---|
| **Correct Posture** |  |  |  |  |
| **Incorrect Posture** |  |  |  |  |
| **Monitoring Parameters** | • Leg Velocity<br>• Kicking Height<br>• Impact Force on the ball | • Separation between Legs<br>• Pressure on the Knees<br>• Single Rep Duration | • Position of the Limbs relative to the center of mass<br>• Body's angular momentum | • Time taken to perform the motion<br>• Alignment of the Back |
| **Decision Models** | • Maintain ideal momentum<br>• Foot too low or too high<br>• Low impact force | • Legs too close<br>• Fast descent<br>• Hunchback posture | • Swayback posture<br>• Feet and Hands not fully extended | • Fast motion<br>• Arching Legs upwards<br>• Back not straight |
| **Customized Pasteables** | • Two Pasteables<br>• Locations: 1 on a Leg & 1 on the kicking Foot | • Three Pasteables<br>• Locations: 2 on the Back & 1 on a Hand | • Four Pasteables<br>• Locations: 2 on the Back, 1 on a Leg, & 1 on a Hand | • Three Pasteables<br>• Locations: 1 on a Leg, 1 on a Hand, & 1 on the Back |
| **Expected Pasteables Cost** | $30 = $20 for Two Base Pads + $10 for CUI sets | $39 = $27 for Three Base Pads + $12 for CUI sets | $49 = $34 for Four Base Pads + $15 for CUI sets | $39 = $27 for Three Base Pads + $12 for CUI sets |

(a)

| | Camera-Based [3], [4] | IMU-Based [5]–[8] | Other Camera-less Systems [9], [10] | Pasteables |
|---|---|---|---|---|
| **Functional Flexibility** | High | Medium | Low | **High** |
| **Ease of Use** | Medium-High | Low-Medium | Low | **High** |
| **External Stationary Reference** | Required | Not Required | Not Required | **Not Required** |
| **Security/Privacy** | Low | High | High | **High** |
| **Affordability** | Low | Low-Medium | Medium-High | **High** |
| **Accuracy** | High | Medium-High | Medium-High | **High** |

(b)

**FIGURE 1.** (a) Different fitness and health applications, along with correct and incorrect postures in a particular activity, monitoring parameters, detection thresholds, and use cases for the proposed Pasteables platform. (b) Comparing various pose and posture detection techniques against the proposed technique built on the Pasteables platform.

in the fitness and athletic domain. It allows on-the-fly hardware component swapping and software reconfiguration using a simple interface. We present the system architecture and design details for Pasteables.

2) We have built multiple prototypes of Pasteables geared toward human pose and posture monitoring that can be configured into an on-body network of sensors.
3) Using these functional prototypes of Pasteables, we have evaluated a set of cameraless pose/posture detection methods and verify them through extensive empirical analysis. This demonstrates the effectiveness and flexibility of the Pasteables platform.

## BACKGROUND AND MOTIVATION

In this section, we provide background on wearables in fitness and health monitoring, as well as existing pose/posture detection techniques. Next, we discuss the need for a reconfigurable and flexible wearable platform for use cases, such as pose/posture detection.

### Wearable Technology: In Fitness and Health Monitoring

Most common wearables are smart patches and smart bands. Smart bandages can treat wounds and other skin injuries as a smart patch by detecting the required antibiotic to prevent the spread of bacterial infections.[11] However, these are often single use and offer little in fitness applications. Smart bands are primarily used in sports training, fitness workout, dance, and physiotherapy as shown in Figure 1(a). Popular devices include the FitBit series and Apple watch series, which cost $80 and $399 for their base models in 2021, respectively.[12,13] There are



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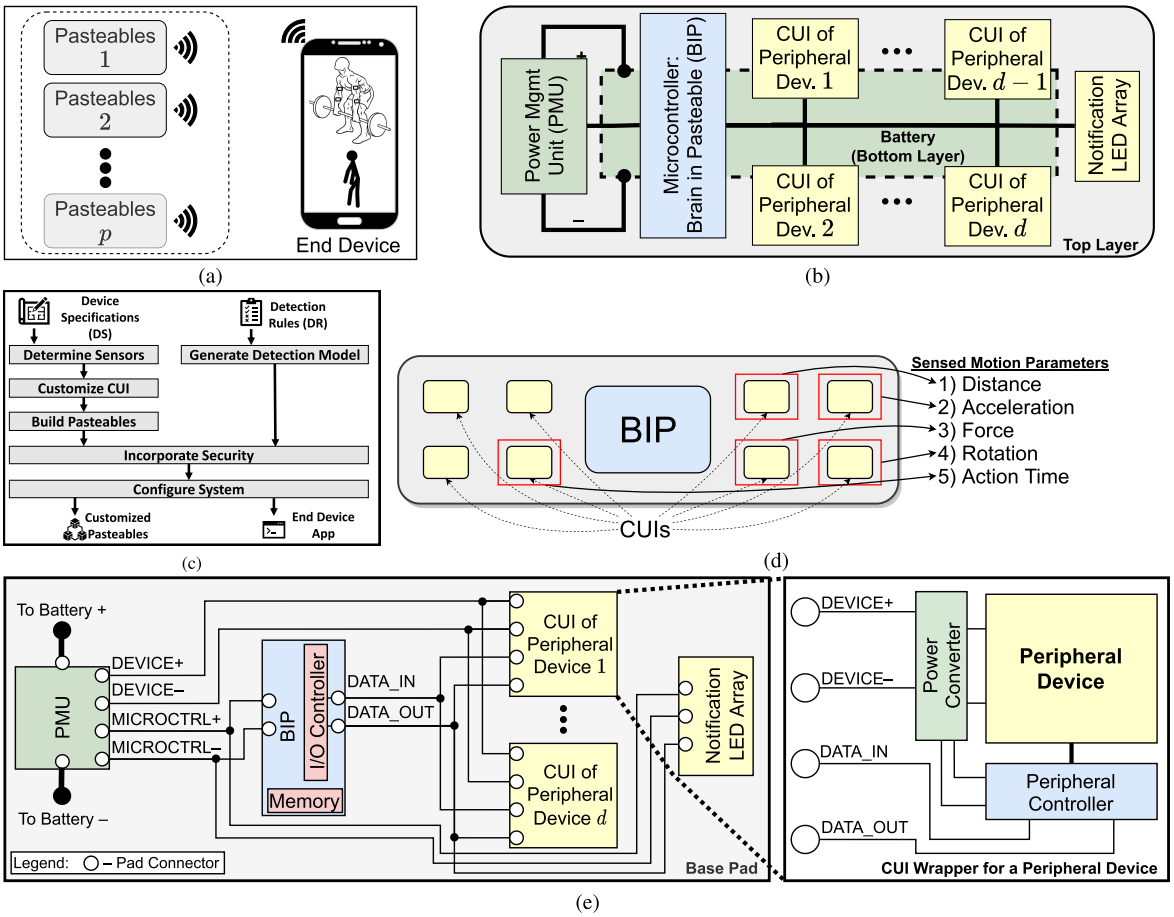



**FIGURE 2.** Proposed Pasteables platform. (a) Connectivity of some Pasteables and an end device. (b) System architecture consisting of: PMU, BIP, CUIs, notification LED array, and a battery. (c) Synthesis process describing the steps for building a system of Pasteables. (d) CUIs with different sensors measuring different parameters used for various fitness and athletic applications. (e) Circuit diagrams of the base pad (left) and a CUI (right), which is a peripheral device in a wrapper circuit.

inexpensive options, but they only have some features of their high-end counterparts. Moreover, these wearables lack customizability in reconfiguring the hardware and software. Switching out sensors and components either is near impossible or requires buying an entirely new system. These limitations restrict the wearer's control and can be costly over time. Thus, an affordable wearable where the wearer can customize its hardware and software should be developed to help solve these issues.

### Existing Pose and Posture Estimation Techniques

Figure 1(b) summarizes pose and posture monitoring approaches in terms of flexibility in detecting different poses/postures, ease of use, need for a stationary reference, security, affordability, and classification accuracy. Camera-based works have explored solutions using visual sensors, which require an external stationary reference where the device location needs to remain constant.[3,4] Mobile devices can implement them accurately and affordably for detecting many exercises. However, cameras pose privacy issues as it could film private data, such as their faces and locations. A cameraless system can help resolve this issue while detecting exercise pose and walking posture and correctness in performing them.

The earliest cameraless system for spinal posture analysis was implemented in Tanaka *et al.*'s


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work,[9] using an electromagnetic (EM) inclinometer. Other cameraless systems used sensors, such as strain gauge, EM, and flex sensors.[10] To create affordable systems, these works used few sensors, which can only detect a few actions. Moreover, an external reference is not needed, as the system's calibration provides the reference. Without cameras, they do not film the user, greatly improving their privacy. Many cameraless approaches use inertial measurement unit (IMU). One such smart patch for back posture detection is Upright Go.[5] Other approaches use motion capturing systems, such as Xsens Dot, LpMoCap, and Noraxon, which require specialized spaces and are prohibitively expensive for the public.[6–8] However, they are significantly accurate for pose and posture detection in these specific environments, but not anywhere else.[10] Thus, an inexpensive, accurate, and easy-to-use cameraless system for detecting various poses can help mitigate issues with existing approaches.

### Need for a DIY and Reconfigurable Monitoring Device

As discussed previously, the existing technologies lack reconfigurability. In addition, designing custom monitoring devices for niche applications is also challenging due to the lack of industry incentive. These limitations must be addressed in newer wearables in order to meet the evolving needs of the fitness and athletics industries. Hence, there is a need for a customizable and easy-to-use wearable platform that can perform various fitness and athletics monitoring, such as pose and posture detection. Such a device can potentially lead to the reduced cost due to reuse and quick/easy development cycles.

## PASTEABLES PLATFORM

We propose Pasteables, a stick-and-peel smart patch wearable with pasted-on reconfigurable hardware components. A group of them can monitor health parameters and limb movements by wirelessly connecting themselves to an end device (e.g., computer and cloud), as shown in Figure 2(a). They can also be configured to create a wireless body area sensor network, with each serving as a reconfigurable edge IoT device. Hence, the wearer can easily set up the hardware on the Pasteables with the software automatically configuring them.

### System Architecture

Figure 2(b) displays the Pasteables hardware platform with two key parts: base pad and pasted-on hardware components. The wearer sticks pasted-on components onto the base pad and base pad onto the skin or clothing. The base pad has top and bottom layers for hardware components and battery, respectively. Pasted-on hardware components, such as a microcontroller called BIP, PMU, and peripheral devices, sit on the top layer. In addition, the battery is on the bottom layer and connects to the Pasteables PMU, powering the platform. The stick-and-peel nature of these Pasteables makes an excellent case for their ease of use, rather than creating a device from scratch.

### Base Pad Circuit

Figure 2(e) shows the base pad and peripheral device circuit diagrams. Within the base pad, the PMU converts the battery's voltage to correct values for the BIP and peripheral devices. BIP initializes, communicates with, and collects data from the pasted-on peripheral devices using its internal I/O controller. It also handles communication with the end device and other Pasteables. An LED array is connected to the BIP, notifying the wearer of their activity status. Like the LED array, the peripheral devices also connect to the microcontroller. With this design, the wearer can customize the hardware within a Pasteables platform by unplugging and replacing components.

### Component With Uniform Interface (CUI)

Peripheral devices (i.e., sensors, actuators, and transducers) are contained in a wrapper circuit to create a CUI, which uses the same material as the base pad. The CUIs are designed to be as bendable as the base pad. A power converter takes the input voltage and shifts it to the correct voltage for the peripheral device if necessary. Moreover, the peripheral controller manages and collects data from the actual


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peripheral device as well as communicates them to the BIP.

### Usage Model and System Synthesis Process

Figure 2(c) depicts the Pasteable designing framework, which helps to create a customized Pasteables system for tracking different fitness and health parameters. It requires both device specifications (DS) and a set of detection rules (DR) for the task. Also, we define a set of measurable entities as $X$. For our current embodiment, we consider $X$ = {distance, acceleration, force, rotation, action time}, as shown in Figure 2(d). $\mathrm{DS} = \{P_1, P_2, \ldots, P_N\}$, where each $P_i \in \mathrm{DS}$ is the $i$th Pasteable's specification. $P_i = \{X_1^i, X_2^i, \ldots, X_{C_i}^i\}$, where each $X_j^i \in P_i$ is sampled from $X$. DR is a set of rules crafted based on values from all the peripheral devices. For example, one rule can be $[\mathrm{angle}(X_1^1, X_1^2) > Th_1]$. The angle between $P_1$ and $P_2$ (computed based on distances $X_1^1, X_1^2$) being greater than $Th_1$ can be one rule for an application. Based on DS, the necessary sensors and their CUIs are generated along with the necessary Pasteables hardware components are crafted. Using DR, the detection model is generated for the given task, along with the necessary security protocols. The final output is the complete system, $S$ = {customized Pasteables, end device app}.

### Self-Awareness of the Pasteables Platform

The uniform and systematic approach to constructing Pasteables helps to create standardized algorithms to self-configure and operate a group of Pasteables and peripherals. RunPasteables() interacts with the network of Pasteables and calls the other procedures to setup and operate the Pasteables. It calls PasteablesSelfAwareness() to autoconfigure the system of Pasteables provided from synthesis framework. To setup the peripheral devices, Algorithm 1.2 is called. After establishing all the Pasteables and peripheral devices, the main procedure collects data using GatherDataFromPasteable() periodically.

## PASTEABLES IMPLEMENTATION

In this section, we discuss our Pasteables prototype design implementation. Then, we talk about the hardware prototype of the base pad and pasted-on components. Next, a prototype implementation of the self-awareness feature is also described.

**Algorithm 1.** Pasteable Operation

```
1: procedure RunPasteables(S, check_time, sample_time)
2:     P = PasteablesSelfAwareness(S)
3:     prev = GetTimeNow()
4:     while True do
5:         curr = GetTimeNow()
6:         if curr − prev ≥ check_time then
7:             P = PasteablesSelfAwareness(S)
8:             prev = GetTimeNow()
           data_vec = []
9:         for each Pi ∈ P do
10:            data_enc = GatherDataFromPasteable(Pi)        } Parallel
11:            data_vec.append(DecryptData(data_enc))        } Execution
12:        X = MakeDecision(data_vec)
13:        LogOutcome(X)
14:        Delay(sample_time)
```

**Algorithm 1.1.** Self-Awareness of a network of Pasteables

```
1: procedure PasteablesSelfAwareness(S)
2:     P = DiscoverPasteables(S)
3:     configs = []
4:     for each Pi ∈ P do
5:         if not AuthPasteable(Pi) then                    } Parallel
6:             IsolatePasteable(Pi)                         } Execution
7:         else                                             } Execution
8:             configs.append(PeripheralsSelfAwareness(Pi)) } Execution
9:     ConfigurePasteables(P, configs)
10:    return P
```

**Algorithm 1.2.** Self-Awareness of Peripherals on a Pasteable

```
1: procedure PeripheralsSelfAwareness(Pi)
2:     D = DiscoverPeripherals(Pi)
3:         for each Di ∈ D do                               } Parallel
4:             if not AuthPeripheral(Di) then               } Execution
5:                 IsolatePeripheral(Di)                    } Execution
6:     Pi.D = D
7:     return MakeConfigString(Pi)
```

**Algorithm 1.3.** Gather Data From a Pasteable

```
1: procedure GatherDataFromPasteable(Pi)
2:     data = CollectData(Pi.D)
3:     Y = PerformAI_Inference(data)
4:     return EncryptData(Y)
```



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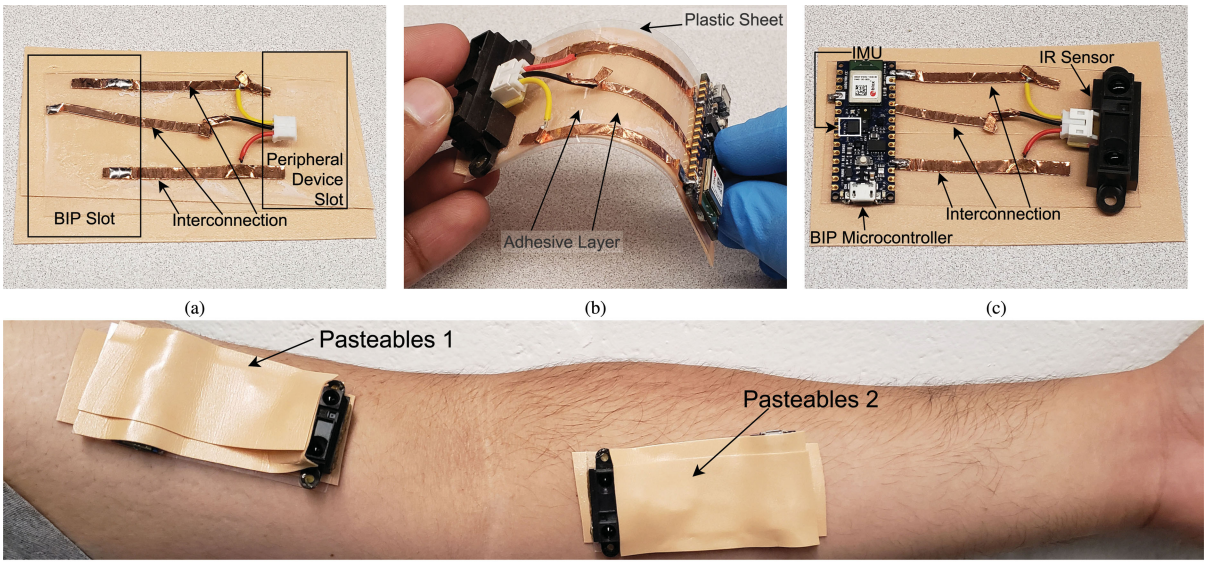



**FIGURE 3.** Pasteables prototype. (a) Copper interconnections are laid out on the base pad, which connects the BIP with the peripheral device. (b) Flexible plastic sheet was added to the adhesive layer, increasing tensile strength of the base pad. (c) Hardware components are pasted on the base pad, showing a complete Pasteables platform. (d) Two Pasteables with an adhesive cover are pasted on the upper and lower arm for fitness monitoring.

### Hardware Prototype of the Base Pad

Figure 3 shows the prototype of the Pasteables platform. The base pad is built on top of a skin-safe adhesive (i.e., Nexcare Waterproof adhesive), as shown in Figure 3(a). It is very flexible, and its tensile characteristics are similar to textile. The BIP and sensor are glued onto the base pad, connected via a removable interconnect using thin copper sheet strips. This interconnect bus is soldered to the BIP and sensor for power and data transmission. It can also contract and expand in unison with the base pad due to added length. A JST PH 2.0 male-to-female connector connects the sensor to the BIP. To improve the tensile strength of the base pad, a thin, flexible plastic sheet was added over the adhesive layer, improving the rigidity slightly, as shown in Figure 3(b). Adding the plastic sheet improves the integrity of the base pad and the adhesive surface; it also maintains the flexibility without needing to apply stickier substance to adhere to the skin or clothing.

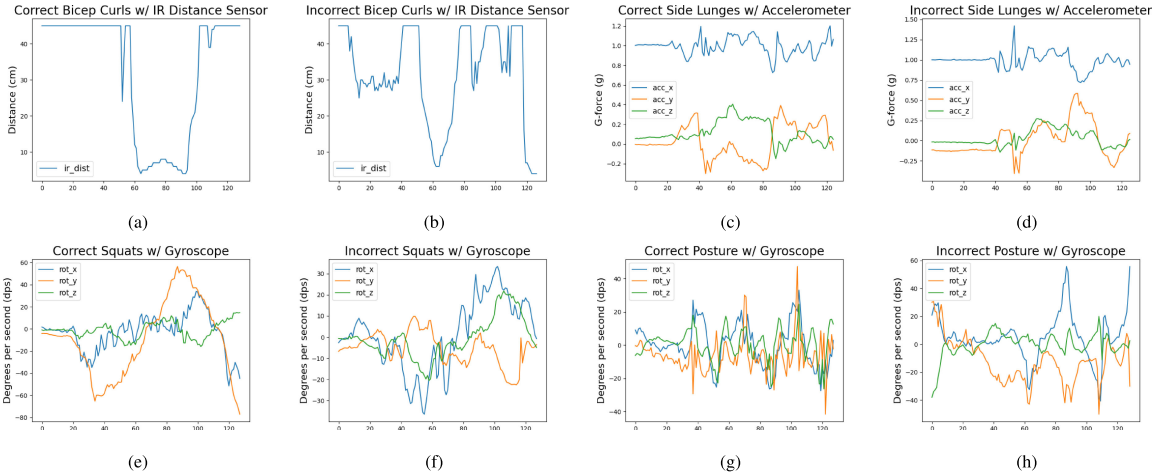
### Device Configuration

Figure 3(c) shows the pasted-on components for this prototype design. The BIP is an Arduino Nano 33 microcontroller due to its small form factor. It also has a Wi-Fi module to communicate with the end device and other Pasteables as well as a simple PMU. The BIP is connected to a 110-mAh lithium-ion battery and provides 3.3 V to the peripheral devices. An IMU and infrared (IR) sensor were chosen for the peripheral devices, i.e., Arduino's IMU and the Sharp GP2Y0A21YK IR sensor. The IMU provides acceleration and rotation data via the accelerometer and gyroscope, and the IR sensor provides the distance measurement. Since they are off-the-shelf components, they have on-board power converters, so an external one is not needed. Using these off-the-shelf components helped simplify the prototype design while maintaining a form factor suitable for on-the-body applications. Future iterations of the Pasteables platform will utilize flexible electronics and PCBs, such as in Biggs *et al.*'s work,[14] to replace all the rigid elements, such as the BIP controller and the IR sensors.

### Software Prototype for the Pasteable Device

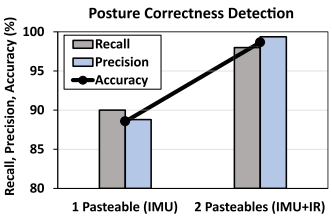
BIP has self-awareness capabilities to autoconfigure and operate a Pasteables platform. Discovering peripheral devices is done by tracking preallocated pins for each device within the BIP. Each Pasteable communicates using their Wi-Fi


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| | Bicep Curls | | Side Lunges | | Squats | |
|---|---|---|---|---|---|---|
| | Setup 1 | Setup 2 | Setup 1 | Setup 2 | Setup 1 | Setup 2 |
| Recall (%) | 83.56 | 91.33 | 91.11 | 89.78 | 100.00 | 97.56 |
| Precision (%) | 83.73 | 93.59 | 86.57 | 92.11 | 100.00 | 97.73 |
| Accuracy (%) | 83.33 | 92.33 | 88.22 | 90.89 | 100.00 | 97.56 |

(i)




(j)

| | Setup 1: One Pasteable (with IMU) | | | | | Setup 2: Two Pasteables (with IMU and IR) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Bicep Curls | Side Lunges | Squats | Standing | Sitting | Bicep Curls | Side Lunges | Squats | Standing | Sitting |
| Recall (%) | 95.11 | 90.44 | 88.44 | 95.78 | 89.33 | 99.33 | 89.78 | 97.33 | 100.00 | 98.22 |
| Precision (%) | 85.01 | 97.70 | 95.06 | 91.31 | 94.39 | 100.00 | 98.40 | 92.87 | 98.61 | 96.23 |
| Accuracy (%) | 91.82 | | | | | 96.93 | | | | |

(k)

**FIGURE 4.** (a)–(h) Sensor data from Pasteables for correct and incorrect exercises/postures. (i) Pose correctness detection for bicep curls, side lunges, and squats with Setups 1 and 2. (j) Posture correctness detection for the same experimental setups. (k) Pose classification for exercises and stationary actions.

modules and the UDP protocol for rapid data transmission. The end device (i.e., desktop computer) configures how to handle the peripherals' data stream based on the Pasteables' configuration strings. The decrypted data packet can be used for training or inferencing, depending on the application requirements. For example, Pasteables can use these data to infer correct pose and posture as well as notify the wearer of the action. Therefore, the software prototype of this Pasteables design seamlessly configures and operates the Pasteables without having the wearer interact with the low-level implementation.

## EXPERIMENTATION

With the Pasteable prototypes, we perform pose and posture detection for multiple health and fitness monitoring applications. We experiment with two different setups for performing three tasks:

1) exercise pose correctness detection,
2) exercise pose classification, and
3) walking posture correctness detection.

The details of these setups are listed in the following.

- Setup 1 has one Pasteable with the IMU as the only sensor.
- Setup 2 has two Pasteables with the IMU and IR sensors.

### Experimental Setup

Five volunteers performed correct and incorrect exercise poses (i.e., bicep curls, side lunges, and squats) and walking postures while wearing


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Pasteables on their body. For example, Figure 3(d) shows Pasteables on the arm. Beforehand, volunteers watched videos from a fitness trainer on the correct and incorrect poses for each exercise.[15] After performing each motion 10 times, the participants performed baseline actions, such as sitting and standing 10 times each. To collect walking posture data, Pasteables were removed and placed on the neck and, later, on the back. Then, the participants walked normally and with a hunched back for 10 times each. We investigate whether using one or two sensors as well as one or two Pasteables is more accurate. Figure 4(a)–(h) shows raw data samples from sensors, which highlight the difference between correct and incorrect poses. All these data points are used to train a random forest (RF) classifier with 500 trees, as it uses many decision trees, making it less prone to outliers. Furthermore, it is validated for 30 folds to collect and verify the classification accuracy, recall, and precision. Thus, this experimental setup provides us with the capability to test different Pasteables configurations and how they affect the accuracy of the pose and posture classification and correctness detection.

### Experimental Results for Exercise Pose Correctness

The first experiment investigates the accuracy of detecting whether an exercise pose is correct using Pasteables. Both correct and incorrect pose raw data, excluding the IR sensor data, were preprocessed using linear discriminant analysis (LDA) technique to reduce the number of features. LDA makes the RF classifier less prone to overfitting, improving the accuracy to above 80%. Figure 4(i) summarizes the first experiment's results. Setup 2 has a mean higher accuracy across the exercises than Setup 1. However, the very high accuracy of the squats pose in Setup 1 could be due to LDA eliminating the noise in the data. The lower accuracy in Setup 2 may be due to more noise introduced when adding more sensors. The IR distance sensor in Setup 2 tracks the distance between the two limbs, which helps to differentiate the correct poses. Consequently, adding more sensors track more physiological parameters, and Pasteables increases accuracy for detecting correct and incorrect exercise poses. Since this platform is affordable to make, attaching more Pasteables and sensors is feasible and can improve the accuracy of the exercise pose classification and correctness detection.

### Experimental Results for Walking Posture Correctness

The second experiment evaluates the accuracy of detecting correct proper walking posture as summarized in Figure 4(j). Like in the first experiment, the posture data were preprocessed using LDA to improve accuracy to above 90%. Setup 2 has a 9% accuracy improvement over Setup 1 compared with the first experiment. Consequently, Setup 2 significantly improves the posture correctness detection like in the first experiment, as the IR sensor tracks the arch of one's spine. With these two experiments, the Pasteables platform also allows for effective pose and posture detection without having a stationary reference, because the Pasteables move with the wearer. Therefore, using multiple Pasteables platforms can perform both detection tasks accurately without this reference.

### Experimental Results for Exercise Pose Classification

The third experiment assesses the nature of the action performed across different exercises as summarized in Figure 4(k). This task is more challenging than the first experiment due to high variance (across individuals and different actions). It looks at the three exercises (bicep curls, side lunges, and squats) along with standing and sitting. Only the correct pose data were analyzed for this experiment. Unlike the first experiment, the pose data were not preprocessed with LDA as accuracies are above 90%. Setup 2 is 5% more accurate than Setup 1, a larger difference compared to the first experiment. This higher accuracy helps differentiate the actions across different exercises using more sensors for monitoring other relevant data. For similar implementations with added reconfigurability, an accurate reconfigurable wearable is feasible for action pose differentiation.

### Extension to Other Fitness and Athletic Applications

Through these three experiments, Setup 2 produced a higher accuracy than Setup 1 for detecting correct pose and postures. As the Pasteables infer the five parameters for posture detection, they



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estimate accurately the correctness of a particular pose or posture. Hence, we believe this can be extended to applications, such as sports training, fitness workout, dance, and physiotherapy, with great accuracy using multiple Pasteables. With the current setup, the cost of each prototype is approximately $30 based on the prototype's bill of materials. Removing unneeded BIP components and accounting for mass manufacturing with scaling, the cost of the base pad would conservatively cost $15 per board and $1–3 per CUI or even less. Compared to the existing consumer wearables, these Pasteables are more affordable and customizable for their intended application.

## CONCLUSION

We have presented Pasteables, a low-cost and hardware/software reconfigurable wearable edge device that can be pasted on body parts, such as band-aids. It can track various pose/postures in diverse domains, such as sports, fitness, physiotherapy, and dance/military training. We have built fully functional prototypes of Pasteables using a flexible plastic base pad, skin-friendly adhesive, and copper strips for communication between different components. We have configured a set of Pasteables for performing cameraless human body pose and posture detection for various fitness applications. The effectiveness of these systems and the flexibility of the Pasteables are quantitatively verified using a set of well-designed experiments using data from multiple subjects. Future work will focus on improving the Pasteables design further in terms of precision and reliability and studying its effectiveness in other applications. To increase the configurability of the Pasteables, the BIP would be developed as a hot-swappable component, such as the CUIs, along with a standardized connection port to the base pad. We believe that Pasteables are promising in many other applications beyond fitness and health, such as dance lessons, appliance monitoring, and human traffic tracking. In the future, we plan to build the base pad using a copper-polyimide sheet for increased flexibility and improved electrical design. Furthermore, we plan to work toward creating a standardized interface between the components and developing algorithms for automatic generation of wrapper circuits for CUIs.

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